

AVYAKT GARG

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EDUCATION

M.S. in Computer Science
University of Wisconsin Madison

Sep 2025 - May 2027
GPA : **4/4**

B.E. in Computer Science and M.Sc. in Biological Sciences
Birla Institute of Technology and Science, Pilani

Nov 2020 - May 2025
GPA : **9.01/10**

TECHNICAL SKILLS

Languages: Python, Go, Java, C/C++, CUDA, JavaScript, TypeScript, SQL, HTML/CSS

Frameworks & Libraries: React, Node.js, FastAPI, SpringBoot, PyTorch, TensorFlow, Scikit-learn, Streamlit, OpenCV

AI & Data: LLMs, RAG Pipelines, Vector Embeddings, pgvector, Approximate Nearest Neighbor Search, PostgreSQL

Infrastructure & Tools: Docker, Git, Kafka, gRPC, REST APIs, SSE Streaming, Agile, JIRA, Mockito, CMake

PROFESSIONAL EXPERIENCE

Uber | *Software Intern*

Jul 2024 - Jun 2025

- Developed an upcoming knowledge work marketplace serving **5+** countries, focusing on data annotation and ML model training
- Implemented a rate-card system in **Java SpringBoot** using MVC architecture and developed **gRPC** APIs for cross-service integration
- Created a **Kafka**-based notification engine with optimized cadence scheduling, improving the customer retention funnel by **3X**
- Built a debug tool with 7 automated checks for **Uber AI Solutions**, reducing developer dependency on recurring issues by **80%**

University of Saskatchewan | *Mitacs Research Intern*

May 2023 - Aug 2023

- Automated data collection via an **RFID** sensor system for a biomedical research lab, saving **3K+** manual hours per year
- Redesigned a **PostgreSQL** database (**6,600+** entries), achieving **90%** faster queries through genealogy-linked relational schema

Indian Institute of Technology, Ropar | *Research Intern*

Jul 2022 - Sept 2022

- Trained and benchmarked **ML** models on **4K+** IoT sensor data points for real-time health monitoring and behavior prediction

PROJECTS

RAG-Powered Portfolio Chatbot | *Full-Stack AI — Go, PostgreSQL, LLMs*

- Built a 3-repo **RAG** system in **Go**: streams **LLM** responses over **SSE**, retrieves context via **pgvector** cosine search on **Voyage AI** embeddings in **Neon PostgreSQL**; pluggable LLM interface supports Gemini and Groq providers with zero-code switching
- Engineered input guardrails (injection detection, topic filtering via cosine distance), query contextualization for pronoun resolution, and **HyDE** (Hypothetical Document Embeddings) for improved retrieval relevance across the knowledge base

GPU-Accelerated Vector Search Engine | *Graduate HPC Project — CUDA C/C++*

- Building an ANN search engine using **IVF-PQ** (Inverted File Index with Product Quantization)—the core algorithm behind **FAISS** and production vector databases powering RAG pipelines, recommendation engines, and large-scale image retrieval
- Implementing custom **CUDA** kernels with shared-memory tiling, **OpenMP** parallel codebook training, and **Thrust/CUB** primitives; benchmarking recall@k vs. QPS tradeoffs on **SIFT1M** with Nsight Compute profiling on an HPC cluster

Graph Learning for Disease Prediction | *Graduate Project — Python, FastAPI*

- Modeled non-i.i.d. clinical relations via Patient Similarity Networks and Bipartite Graphs using **GraphSAGE** and **Node2Vec**
- Achieved 83.5% accuracy and 0.913 AUC-ROC with Nested Cross-Validation; deployed risk assessment app via **FastAPI** and **Streamlit**

Deep Learning Framework for prediction of Intracranial Pressure (ICP) using OCT Scans | *Prof. S. Raman*

- Built a 3D ResNet-18 classifier for intracranial pressure prediction using Optical Coherence Tomography(OCT) retinal scans
- Preprocessed 512-frame videos to isolate key regions of interest and fine-tuned architecture for safe patient threshold prediction

RELEVANT COURSES

High Performance Computing (CUDA/GPU), Machine Learning, Artificial Intelligence, Data Structures & Algorithms, Database Systems, Operating Systems, Computer Networks, Human Computer Interaction

ACADEMIC ACHIEVEMENTS

Globalink Research Scholarship — MITACS, Canada: Selected out of 30k+ applicants for research internship

Feb 2023

Scholarship Awardee — AWaDH, Govt. of India: For AI work in agriculture domain

Apr 2022

Gold Medalist — Delhi Public School R.K. Puram: 9 years of academic excellence

May 2019

86th Rank — JSTSE — Directorate of Education, Delhi: Science merit scholarship

Feb 2017